

WASP-11b

R-0898

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_171226 · created 2026-07-29T03:12:57+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458113.63845 +/- 0.00095 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1247 +/- 0.0038
Transit depth (Rp/Rs)^2	1.56 +/- 0.095 [%]
Orbital Inclination (inc)	88.82 +/- 0.48
Airmass coefficient 1 (a1)	1.1056 +/- 0.0074
Airmass coefficient 2 (a2)	-0.097 +/- 0.012
Scatter in the residuals of the lightcurve fit is	0.67 %
Best Comparison Star	None
Optimal Aperture	4.54
Optimal Annulus	9.88
Transit Duration (day)	0.1036 +/- 0.0023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1247 ± 0.0038 vs 0.131 (deviation 1.14σ)
Duration fit vs published	0.1036 d vs 0.1075 d (deviation 1.16σ)
Mid-time O–C	-3.9 min
Depth SNR	22.5
Residual scatter	0.67 %

Light curve

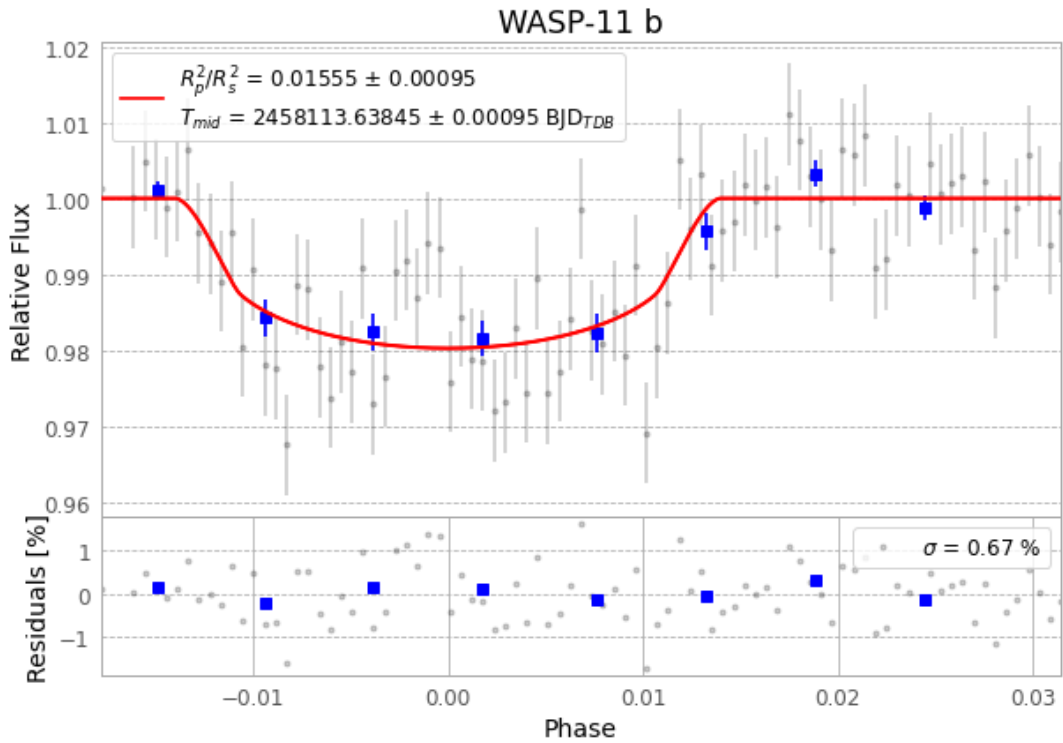


Figure 1 — FinalLightCurve_WASP-11 b_2017-12-25.png (SHA-256 5075b84455fd9fcb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [364, 249], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 89331df490c616b6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

87 FITS frames (57 MB), HATP-10171226013612.FITS → HATP-10171226060012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-171226.log (f376b9bdacf1...), FinalLightCurve_WASP-11 b_2017-12-25.png (5075b84455fd...), FinalLightCurve_WASP-11 b_2017-12-25.csv (6e93cc1f4941...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 03:12Z.